ZHIYAN LI

lzy001Yuki@gmail.com GitHub Homepage

EDUCATION

Shanghai Jiao Tong University, B.Eng. in Computer Science Sep. 2023 – June 2027 (expected)

- **ACM Honors Class** in Zhiyuan College, an elite research-oriented CS program (top 5% of students).
- **GPA: 89.2/100**
- **Selected courses:** Computer Vision (**100**), Data Science (**98**), Information Theory (**95**), Large Language Models (**94**), Compiler Project (**94**), Introduction to Physics II (Honors) (**92**), Machine Learning (**92**).

RESEARCH INTERESTS

- Multimodal perception and generalization in robot learning, especially tactile sensing.
- Lifelong robot learning and generative memory.
- Test-time adaptation using steering policies and agentic methods.

RESEARCH EXPERIENCE

Visiting Student, Harvard University June 2026 – Present
Embodied Minds Lab, advised by Prof. Yilun Du;
Harvard AI and Robotics Lab, advised by Dr. Mengyu Wang.

Research Intern, MVIG-RHOS Lab, Shanghai Jiao Tong University Began June 2025
Co-advised by Prof. Cewu Lu and Prof. Yonglu Li.

Visiting Student, Tsinghua University June – July 2025
Institute for Interdisciplinary Information Sciences (IIIS).

PUBLICATIONS

* Equal contribution; † Equal supervision.

GEMA: Generative Memory for Robotic Manipulation
Zhiyan Li, Pembe Gizem Ozdil, Ben Zandonati, Ruojin Cai, Hang Le, Tomás Lozano-Pérez, Leslie Pack Kaelbling, Weirui Ye†, Yilun Du†.
Under review, CoRL 2026 Workshop on Memory for Robot Foundation Models.

SteerAct: Temporally Gated Single-Feature Steering of Frozen Action Models
Kaichen Zhou*, **Zhiyan Li***, Chongkai Gao, Yuhan Wang, Wenlong Huang, Maggie Wang, Fangneng Zhan, Manling Li, Qianqian Wang, Mac Schwager, Zhuang Liu, Li Fei-Fei, Mengyu Wang†, Paul Pu Liang†.
Under review, ICLR 2027.

VISTA: A Multimodal Egocentric Benchmark for Goal-Oriented Perception and Assistance to Blind and Low-Vision Users
Noé Mosca*, Mohamed Rayan Barhdadi*, Muhammad Hamza*, Joshua Yeh*, Suryodaya Bikram Shahi*, **Zhiyan Li**, Selim-Antoine Lali, Weiyue Li, Hasan Kurban, Erchin Serpedin, Gang Luo, Yue Zhang, Minghan Li, Mengyu Wang.
Under review, ICLR 2027.

TactiLDM for VLA: Learning Unified Visual-Text-Tactile Representations to Empower Vision-Language-Action Policies
Zhiyan Li*, Hantao Jiang* Nov. 2025 – Jan. 2026
Preprint · Code

INDUSTRY EXPERIENCE

Research Intern, Huawei 2025
Central Software Institute; research on multi-agent systems.

RESEARCH PROJECTS

Decoformer *Python, Core Researcher* May 2025

· Replaced CrossFormer's cross-time attention stage with DLinear decomposition for long-term forecasting, achieving **20% higher accuracy** and **1.25× faster inference**.

MxCompiler *Java, Core Researcher* Oct. 2024

· Implemented an Mx compiler with semantic analysis, RISC-V code generation, and optimizations including **Mem2Reg**, **dead code elimination**, **sparse conditional constant propagation**, and **global-to-local variable promotion**.

RISC-V CPU *Verilog, Core Researcher* Oct. 2024

· Designed a five-stage RV32IC processor with Tomasulo-based out-of-order execution; validated it through simulation and FPGA deployment.

HONORS AND AWARDS

- **Foresight-Sequoia Talent Development Fund** 2025
  Awarded to 5 outstanding students in the ACM Class per year.
- **Zhiyuan Honors Scholarship** 2023, 2024, 2025
  Recognition for the top 5% of SJTU students annually.
- **Shanghai Jiao Tong University Merit Scholarship** 2025
- **Provincial Second Prize**, China Undergraduate Mathematical Contest in Modeling 2024
- **Provincial Second Prize**, The Chinese Mathematics Competition 2024
- **Silver Medal, National Second Prize**, 39th Chinese Physics Olympiad 2022

ACADEMIC SERVICE

Junior Organizer and Reviewer Since Sep. 2026
1st Workshop on Physical World AI (PhysWorldAI), NeurIPS 2026

TEACHING EXPERIENCE

Principles and Practice of Computer Algorithms, **Teaching Assistant** Summer 2025
Data Structures (Honors), **Teaching Assistant** Spring 2025

SKILLS

Languages	Mandarin Chinese (native); English (CET-6: 610; speaking: Excellent)
Programming	Python, MATLAB, C++, Java, Verilog
Tools	LaTeX, Markdown, Git, ROS 1/2